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SUMMARY

Principal Software Engineer with 25+ years spanning .NET/backend development, AI agent infrastructure and governance, cloud, and security. Recently building production AI agents and Model Context Protocol (MCP) servers, architecting ML pipelines for an AI startup, and maintaining a security-hardened fork of the NSA's Ghidra reverse-engineering platform—and architecting Scylla, a ground-up hexagonal reverse-engineering platform built on it. Deep C#/.NET and Python background, with a track record of clear technical writing and developer enablement. USMC veteran.

SELECTED OPEN-SOURCE & AI PROJECTS

GayHydra — hardened fork of NSA Ghidra

C++, Java

- Sole architect of a security-hardened fork of the NSA's Ghidra reverse-engineering platform (a multi-million-line C++/Java codebase), driving a 42-item principal-architect audit into shipped releases across security, decompiler, and CI/release engineering.
- Hardened the Java deserialization attack surface (default-reject class filtering; bounds-checked packed-file decode that rejects forged / zip-bomb inputs) and modernized the decompiler's host-worker IPC into a versioned, schema-validated protocol.
- Stood up a multi-OS GitHub Actions pipeline producing Sigstore/cosign keyless-signed release artifacts with a CycloneDX SBOM, on a dual-remote (GitHub + Codeberg) release workflow.

Scylla — hexagonal reverse-engineering platform on the GayHydra engine

Rust, Cap'n Proto, gRPC, MCP

- Architected and built a ground-up, ports-and-adapters reverse-engineering platform that wraps GayHydra's engine as a durable Rust core behind disposable protocol adapters (an MCP server for AI agents first), with the RE domain model serialized as a stable, zero-copy Cap'n Proto artifact — the “narrow waist” that decouples every producer (analysis engine) from every consumer (UIs, agents, collaborators).
- Locked 40+ architecture decisions with written rationale: bet on the slowest-moving layer (the RE domain model) and relocated volatility to sheddable edges, so the analysis engine, the wire protocol, and the agent adapter can each be replaced without touching the core.
- Designed a four-pass identity-anchored matcher (exact signature, architecture-independent string/import/call anchors, reciprocal-best fuzzy, and call-graph propagation) that carries analyst annotations across re-analysis, recompilation, and *cross-architecture* rebuilds (x86-64, aarch64, and 32-bit) under a hard “zero silent mis-attachment” invariant — enforced by a ratcheted release gate over an 11-class binary corpus, so low-confidence matches degrade to flagged-for-review, never a wrong write.
- Built the engine as a sandboxed gRPC service with a warm co-resident GayHydra runtime (roughly 3x faster than cold per-call analysis) plus a worker pool for concurrency and a cold-subprocess fallback behind the same RPC; de-risked the major bets with empirical spikes before committing the build (proving Ghidra Version Tracking the wrong tool for cross-architecture and a Go-aware producer the right one, by measurement).
- Contained the adversarial-binary blast radius: the parser runs in a no-egress container (no network namespace; gRPC over a bind-mounted Unix socket) behind a documented seam-by-seam threat model, a bounds-checked total artifact loader, and keyless Sigstore signing of the CLI and engine image.

KaliMCP & PerplexityAgent

Python

- Two security-hardened Model Context Protocol (MCP) servers exposing, respectively, a curated subset of Kali Linux security tools and the Perplexity Search/Sonar APIs to AI agents; per-invocation refuse-list guards and audit logging, hardened to NSA MCP security guidance.

omind

Python

- Harness-agnostic memory & governance layer that gives every AI coding agent on a machine one shared, file-based long-term memory and a single policy-enforcement point — wired across Claude Code, Codex, and other harnesses so a rule learned under one enforces under all, with one-command reproducible setup and a local web app to browse and edit agent memory.
- Designed the “checks and balances” that keep agents honest: a per-turn gate that blocks tool use until the agent consults its memory, a relevance verifier that rejects off-topic consults (so the gate can’t be cleared by reading an irrelevant note), deliberate opt-in for high-risk actions (e.g. publishing to a public remote), and a compliance audit trail of violations, repeat offenders, and verifier verdicts — all fail-open, so a broken hook never wedges the agent, with self-healing drift detection and a doctor that diagnoses and repairs the wiring.

Open-Source Contributions

C#/.NET, Python, Java

- **Omnigent** (Python) — contributor to an open-source meta-harness for AI agents (a policy/sandbox layer over Claude Code, Codex, and custom agents); authored three merged P0 security fixes (PRs #158, #161, #163) hardening session input validation (RCE), OIDC login (identity spoofing), and policy fail-closed enforcement.
- **Libation** (C#/.NET) — contributor to the open-source Audible library manager; merged PR #1814 (locale-dependent unit-test fix) with additional feature PRs in review.
- **hermes-agent** (Python) — contributor to an open-source personal AI agent framework with persistent memory, scheduled autonomy, and multi-provider LLM routing with automatic fallback.
- **OWASP ZAP** (Java) — ongoing contributor to the OWASP flagship web application scanner.

EXPERIENCE

Ronin 48

Mar 2026 - Present

Principal Software Architect & ML Engineer (Contract)

Remote

- Designing overall system architecture for an AI startup; defining technical standards, guiding technology selection, and ensuring scalable, production-ready software design across the platform.
- Implementing and operationalizing machine learning pipelines; guiding model selection and deployment strategy, and integrating ML capabilities into production systems.

Career Break

Nov 2024 - Mar 2026

EMT-B Certification & Community Service

Minden, Nebraska

- Completed Emergency Medical Technician (EMT-B) certification and joined Minden Volunteer Fire Department as probationary firefighter / EMT. Self-directed study in AI/ML systems.

Black Hills Information Security / Active Countermeasures

Mar 2024 - Nov 2024

Senior Developer

Remote

- Upgraded Windows Service from dotNET Framework 4.x to dotNet 8, refactored code to be testable and created unit tests. Built Windows ports of Unix software. Wrote documentation and tutorials for blog posts and webcasts.

CrowdStrike

Aug 2022 - Mar 2024

Senior Technical Support Engineer

Remote

- SME for SaaS and on-prem LogScale(Humio), Kafka, and log shippers. Created sandbox infrastructure using terraform and k8s. Wrote automation and tooling in bash and Python.

CloudHesive

Dec 2021 - May 2022

Cloud Operations Supervisor

Remote

- Led the Cloud Operations team supporting 2000+ customer systems across AWS, Azure, and VMware; hired and developed engineers focused on security and operational excellence.

CloudHesive

Aug 2021 - Dec 2021

Cloud Support Engineer II

Remote

- Supported more than 2000 customer systems on AWS, Azure, and VMWare Infrastructures. Used Datadog, SumoLogic, and NewRelic to monitor customer systems and networks. Managed backup and recovery processes. Technologies included CloudFormation, CloudWatch, EC2, EFS, ELB, IAM, RDS, S3, VPC, and Workspaces.

Tmutla

Feb 2021 - Aug 2021

Cybersecurity Consultant

Grand Island, Nebraska

- Performed Web Application Penetration tests and wrote customer reports.

Reinke Manufacturing Company

Dec 2015 - Feb 2021

Senior Developer

Remote

- Led application development of middleware connecting an AS/400-based ERP system and a WinForms sales application. Built the authentication/authorization system using ASP.NET MVC 4. Stood up and maintained Azure infrastructure. Evaluated 3rd-party SAST tools and made recommendations for their implementation.

Alpine Testing Solutions, Inc.

2014 - 2015

Software Developer

Remote

- Monitored and mitigated data import errors; created T-SQL queries for data imports and exports; wrote XML for manual data imports.

Orthman Manufacturing, Inc.

2012 - 2014

IT Specialist II

Lexington, Nebraska

- Led virtualization project to migrate physical Windows Machines to virtual hosts. Created .NET Applications for internal use.

Intellicom Computer Consulting, Inc.

2009 - 2012

Senior Engineering Technician

Kearney, Nebraska

- Installed and maintained Active Directory, File, and Exchange servers.

Earlier Experience

1998 - 2008

- **United States Marine Corps** – Information Specialist (2003–2008), Washington, DC. Operated and supported the USMC network; graduate of the USMC Information Assurance Manager Course.
- **Rackspace** – Linux Systems Administrator (2003), San Antonio, TX.
- **Cimarron Software Services** – Systems Analyst at NASA Johnson Space Center (2001–2002), Houston, TX.
- **MegaHaus** – Help Desk Technician (1998), Dickinson, TX.

TECHNICAL STRENGTHS

AI/ML	Model Context Protocol (MCP) servers, agentic tooling & orchestration, tool-use / function calling, Claude Code & Codex, LangChain, LangGraph, Hugging Face, OpenAI & Anthropic APIs, Ollama, RAG pipelines, vector databases (pgvector, Pinecone), prompt engineering
Cloud	AWS (EC2, S3, RDS, IAM, VPC, ELB, EFS, CloudFormation, CloudWatch), Azure, Terraform
Tools	Burp Suite, Postman, Wireshark, NMap, LogScale (Humio), Kafka, Docker, Kubernetes
Languages	Python, C#, TypeScript, JavaScript, VB.NET, T-SQL, PowerShell, Bash
Certifications	SSCP, AWS Solutions Architect Associate, AWS Cloud Practitioner, CompTIA A+, CompTIA Network+, MS: Programming in C#
Memberships	OWASP, ACM, DC402, RITSEC, SPARSA.ORG, OperationCode

EDUCATION

Eastern University MS, Data Analytics (In Progress)	2026 – Present
University of Maryland University College Bachelor of Science, Computer and Information Science	
Rochester Institute of Technology MicroMaster's in Cybersecurity	
Kennesaw State University Graduate Certificate, Computer Science	
Colorado State University Graduate Certificate, IT Project Management	
University of Maryland University College Undergraduate Certificate, Unix System Administration	

VOLUNTEER EXPERIENCE

Community & Industry Service	2021 - Present
· Minden Volunteer Fire Department — Probationary Firefighter / EMT-B (Dec 2024 – Present), Minden, NE. · Women in CyberSecurity (WiCyS) — Student Mentor (2021–2024). · Harvard University — Teaching Assistant, Online Computing Mini Course (Spring 2022). · Army National Guard — Red Team (OPFOR) Lead, Cybershield 2021.	